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Date
30.11.2021

Reference	Application No./Patent No.
EUR1621WEP	19898944.4 - 1001 / 3872534 PCT/KR2019014018
Applicant/Proprietor Ftlab Co., Ltd.	

Communication

The extended European search report is enclosed.

The extended European search report includes, pursuant to Rule 62 EPC, the supplementary European search report (Art. 153(7) EPC) and the European search opinion.

Copies of documents cited in the European search report are attached.

☒ 0 additional set(s) of copies of such documents is (are) enclosed as well.

Refund of the search fee
If applicable under Article 9 Rules relating to fees, a separate communication from the Receiving Section on the refund of the search fee will be sent later.

Should you wish to further prosecute this application in the examination phase, your attention is drawn to the provisions of Rule 70a EPC. An invitation to respond to the extended European search report will be issued shortly (R. 70(2) EPC).

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SUPPLEMENTARY EUROPEAN SEARCH REPORT

Application Number
EP 19 89 8944

DOCUMENTS CONSIDERED TO BE RELEVANT

Category	Citation of document with indication, where appropriate, of relevant passages	Relevant to claim	CLASSIFICATION OF THE APPLICATION (IPC)
X Y	JP H08 136660 A (ALOKA CO LTD) 31 May 1996 (1996-05-31) * paragraphs [0012], [0020], [0021], [0023], [0028] - [0036], [0042] - [0045], [0048], [0049]; figures 1,2 *	1,12,26 2-11, 13-25,27	INV. G01T1/178 G01T1/18
Y A	JP 2012 002736 A (SUMITOMO HEAVY INDUSTRIES) 5 January 2012 (2012-01-05) * paragraphs [0031], [0033]; figures 1-3 * *	11,16 14	
Y	KR 101 583 056 B1 (KOREA ATOMIC ENERGY RESEARCH INSTITUTE) 7 January 2016 (2016-01-07) * paragraphs [0012], [0040]; figure 1 *	13	
Y	US 10 132 936 B2 (FTLAB CO LTD [KR]) 20 November 2018 (2018-11-20) * column 4, line 1 - column 5, line 38; figures 1,4,5 * * column 8, lines 15-22 *	2-10,14, 15, 17-25,27	

TECHNICAL FIELDS SEARCHED (IPC)

G01T
G21F

The supplementary search report has been based on the last set of claims valid and available at the start of the search.

Place of search	Date of completion of the search	Examiner
The Hague	24 November 2021	Van Ouytsel, Krist'l

CATEGORY OF CITED DOCUMENTS

X : particularly relevant if taken alone
Y : particularly relevant if combined with another document of the same category
A : technological background
O : non-written disclosure
P : intermediate document

T : theory or principle underlying the invention
E : earlier patent document, but published on, or after the filing date
D : document cited in the application
L : document cited for other reasons

& : member of the same patent family, corresponding document

EPO FORM 1503 03.82 (P04C04)

ANNEX TO THE EUROPEAN SEARCH REPORT ON EUROPEAN PATENT APPLICATION NO. EP 19 89 8944

This annex lists the patent family members relating to the patent documents cited in the above-mentioned European search report. The members are as contained in the European Patent Office EDP file on
The European Patent Office is in no way liable for these particulars which are merely given for the purpose of information.

24-11-2021

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
JP H08136660 A	31-05-1996	NONE	
JP 2012002736 A	05-01-2012	NONE	
KR 101583056 B1	07-01-2016	NONE	
US 10132936 B2	20-11-2018	CN 107257932 A	17-10-2017
		EP 3252505 A1	06-12-2017
		KR 20170023599 A	06-03-2017
		US 2018031712 A1	01-02-2018
		WO 2017034158 A1	02-03-2017

Information on Search Strategy - Pilot phase (see OJ 2015, A86)
The type of information contained in this sheet may change during the pilot for improving the usefulness of this new service.

Application Number
EP 19 89 8944

TITLE: APPARATUS FOR MEASURING RADON AND THORON BY USING IONIZATION CHAMBER

APPLICANT: Ftlab Co., Ltd.

IPC CLASSIFICATION: G01T1/178, G01T1/18

EXAMINER: Van Ouytsel, Krist'l

CONSULTED DATABASES:

CLASSIFICATION SYMBOLS DEFINING EXTENT OF THE SEARCH:

IPC:

CPC: G01T1/167, G01T1/161, G21F9/02, G01T1/178, 2G188/BB06 /CC01 CC02 DD10 HH09

FI/F-TERMS:

KEYWORDS OR OTHER ELEMENTS FEATURING THE INVENTION:
An apparatus for measuring radon and thoron using an ionization chamber, comprising: a pump for air inflow suctioning and sending external air to at least one channel; a first sensor module receiving the air outputting an alpha particle detection signal of particles having a predetermined size; an air inflow delay module receiving the air for inflow through a second gas line in a second channel; a second sensor module and a control module receiving in real time alpha particle detection signals output from the first and second sensor modules, comparing and analyzing waveforms with signal pattern information data according to external noise signal types stored in advance, discriminating normal or abnormal alpha particle detection signals, counting the normal alpha particle detection signals discriminated for a predetermined measurement time, and calculating radioactive ray concentration values on the basis of the counted number of times of the normal signals.

The examination is being carried out on the **following application documents**

Description, Pages

1-36 filed with entry into the regional phase before the EPO

Claims, Numbers

1-27 filed with entry into the regional phase before the EPO

Drawings, Sheets

1/7-7/7 filed with entry into the regional phase before the EPO

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 JP H08 136660 A (ALOKA CO LTD) 31 May 1996 (1996-05-31)

D2 JP 2012 002736 A (SUMITOMO HEAVY INDUSTRIES) 5 January 2012 (2012-01-05)

D3 KR 101 583 056 B1 (KOREA ATOMIC ENERGY RESEARCH INSTITUTE) 7 January 2016 (2016-01-07)

D4 US 10 132 936 B2 (FTLAB CO LTD [KR]) 20 November 2018 (2018-11-20)

1. Inventive step

The present application does not meet the requirements of Article 52(1) EPC because the subject-matter of claims 1-27 does not involve an inventive step within the meaning of Article 56 EPC.

1.1 Claim 1

Document D1 (fig.s 1,2) discloses:

an apparatus (fig.s 1,2, ref.s 26,28,30) for measuring radon and thoron (par.[21]) using an ionization chamber (36,136), the apparatus comprising:
air inflow suctioning (34b,134b, par.[30]) and sending external air to at least one channel (38,138);
a first sensor module (136) receiving the air for air inflow through a first gas line in the

first channel (134b,138), and outputting an alpha particle detection signal of an electrical signal by detecting alpha (fig.2, par.[12]) particles discharged from radon and thoron included in the air flowing (alpha particles) in an ionization chamber having a predetermined size (136);
an air inflow delay module (filters 32, apparatus 26) receiving the air for air inflow through a second gas line in a second channel (34b,38), delaying the air for a predetermined delay time (par.s [28-29]), and then outputting the air (fig.2, ref.38, par. [30]);
a second sensor module (36) receiving air output from the air inflow delay module and outputting an alpha particle detection signal of an electrical signal by detecting alpha (fig.2, par.[12]) particles discharged from radon and thoron (*thoron mostly absorbed by filter*) included in the air flowing in the ionization chamber having a predetermined size (36); and
a control module receiving in real time (par.[49]) alpha particle detection signals of electrical signals output from the first and second sensor modules (fig.1, par.s [42-45]), comparing and analyzing waveforms with signal pattern information data according to external noise signal types stored in advance, discriminating normal or abnormal alpha particle detection signals, counting the normal alpha particle detection signals discriminated for a predetermined measurement time, and calculating radioactive ray concentration values on the basis of the counted number of times of the normal alpha particle detection signals (par.s [42-45]).

1.1.2 The subject-matter of claim 1 seems to differ from this known apparatus in that a pump is used for air inflow suctioning and sending on.

D1 does not specify this but states that the air is naturally evacuated and refers to the prior art systems using a pump.

1.1.3 The problem to be solved therewith would thus appear to be: how to regulate the the flow rate of the air intake and sending on.

1.1.4 The solution: to use a pump, is then, as D1 mentions already, known in the art and cannot be seen to bear any inventive step. Also ion chambers open to external atmosphere are well-known in the art and D1 (par.[32]) discloses the pressure in the chamber to be about 1atm, allowing the skilled person to easily modify the air inflow to the chamber as claimed on the basis of ordinary laboratory practise.

The claim is therefore seen to lack an inventive step against D1 alone.

1.2 claim 16: document D2 (fig.1, par.[21]) also discloses the use of a valve system. The skilled man, starting from D1 would know the obvious tools to control the air flow, such as in D2, and therewith arrive at the subject-matter claimed.

1.3 Dependent claims

claims 2-10 and 17-25: document D4 (fig.s 1,4,5, col.4, line 1 to col.5, line 38) e.g. discloses the additional features, is in fact the applicants own prior art. The claims are therefore seen to lack an inventive step against a combination of D1 with D4.

claims 11,12: document D2 (fig.s 1-3, par.s [31,33]) discloses the air inflow delay module comprising multiple stacked cases, with openings therebetween, further to provide partitions is seen to be an obvious measure known in the art. A combination of D1 with D2 is therefore seen to take away any inventive step of the claims.

claim 13: document D3 (fig.1, par.s [12,40]) discloses the wound air pipe in the shape of a circular spring, a combination of D1 with D3 is therefore seen to take away any inventive step of the claim.

claims 14,27: D1 (par.s [20,28]) discloses a delay time of 10 mins; whilst D4 (col.8, lines 15-22) specifies the measurement time as claimed. Using a pump for the inflow and exhaust of air is seen to represent an obvious measure well-known in the art.

claim 15: D1 combined with D4 also discloses the signal processing as in claim 15.

claim 26: document D1 (fig.1, par.s [42-45]) discloses giving out 4 concentration values.

2. Clarity

The application does not meet the requirements of Article 84 EPC, because claims 1 and 15 are not clear.

claims 1 and 15: in particular, reference is made to comparing and analyzing waveforms, but these waveforms are not related to the signals output from the first and second sensor modules, a link should be introduced.